Book Reading Assignment

Part A

I. Modeling and Simulation in Python – Allen Downey

1. I recommend this book for people who are very new to programming and started coding only with python, because book as very detailed information starting from basic coding to solving equations from the lab experiments data. I have learned "modeling frame work: Abstract, Analysis and simulation and Validation" are the three components of coding.

2. Did not demonstrated an example that covers all the topics, just provided individual examples for each topic. While trying to solve the example that has explained in the Friday's class faced difficulties.

II. A Students Guide to Python for Physical Modeling – Kinder & Nelson

1. This book is particularly telling how the mathematical expressions works in python. So, it is for people who has basic knowledge in programming and gives insights to solve mathematical problems.

2. I have tried solving np.sqrt(2) by importing numpy as np.float64(1.41421…), I have searched in the for elimination np.flot64, but couldn't find, finally Claud explained what is np.float64 and elimination process of that. Additionally, the book does not tell how to get an integer answers or required number of decimal answers.

III. Think Python – Allen Downey

1. This is useful for everyone who is using python for programming, because similar to Allen Downey's Modelling and Simulation in Python book, this book also explained all the topics that are required for basic coding particularly, using of variable and statements. Addition to this, he has explained the debugging topic by mentioning different types of errors and reading of those errors.

2. He has mentioned the types of errors during the programming, but failed to provide the detailed way of over coming those errors. I felt it would be more helpful for people very new programming, if had demonstrated debugging with examples and also failed to differentiate the usage of numpy and math.

IV. Scientific Python Lectures – Community Written

1. This book highly recommended for people like me, who thinks why python is required for solving mathematical problems despite having MATLAB-that is much friendly to solve complex problems when compared to programming languages.

2. Every topic in this book is only compared with the other programming languages not exactly provided the detailed information about the topics. Particularly, the topics are directly started from the solving mathematical expressions and plotting the data.

V. The Python Tutorial – Documentation

1. This book is for people who are good with programming in other languages and trying to transfer their skills to python because it focused on advanced topics like data structures and modules.

2. Despite tried to understand the topics from the examples but failed to do because it has not provided why the particular topic is used like in flow statements- the range() function, there is only example not provided why it is used and how to correct the errors.

VI. Introduction to Programming with Python – David Malan

1. This book is perfectly good for people who has chemical engineering skills but lack of coding skills. This was fully designed starting from functions, variables conditions and loops that helps to code.

2. I have tried one of the examples that focus on the structure got an error, was unable to over come that due to unavailability of the topics.

Part B

VII. Python Crash Course – Eric Mattes

1. This book is helpful for everyone those who are very new to programming because unlike other books this book was perfectly focused on the basic programming topics and explained very detailed by framing real examples.

2. This is limited to only understating and learning the basics in coding and way of programming, might not help to solve chemical engineering models. Even in the projects at last of the booked, fully focused on the examples that are not related to mathematical expressions.

VIII. Fluent Python – Luciano Ramalho.

1. I recommend this book for people exceptionally good at programming and now planning to transfer their knowledge and gain the skills in python as well, because the topics in this book was fully focused on the fully immersive programming.

2. Tried to code "show_count" by using functions. I was able to do the coding but did not get how these specific functions: show_count, mypy, pytest are being used.

IX. Pyse: Python Computations in Science and Engineering- John Kitchin.

1. This is one of the best books I have read from the part A, Part B and Extra books posted. This book is helpful for all the people with irrespective of programming knowledge, because people who have basic knowledge in programming can learn detailed coding and people who are good at programming can learn modeling chemical engineering problems by python.

2. Despite stating detailed information on programming and using chemical engineering examples to demonstrate, it was failed to explain the python basics like: calling variables and list and using of loops.

Part C

X. W3Schools Python – Tutorial

1. This book is similar to 'python crash course' and Allen downey's two books. Topics in the books are started from the very basic of programming and provided two small examples, that helps for people who are very new to python.
2. Despite having explained many topics in python but failed to relate it to the mathematical or chemical engineering related problems. This might not fully helpful for this course, only helpful what type of loops and functions to be used to write the code.

Summary:

All the 10 books are explained the programming perfectly and these are useful for coding in their ways. But, among these 10, Modeling and simulation in python and pyse are best resources to learn programming as well as modeling chemical engineering problems by coding. Because, Modeling and simulation book helps to learn the basics of programming and types of language components used, and pyse helps to relate these basic and solve chemical engineering problems by coding. So, during this course I will be following these two books for programming.